Majid Haji Bagheri

PhD Candidate · Nanotechnology · Energy Harvesting & Storage · Device Physics · MEMS / Nanofabrication

Electrical & Computer Engineering, Waterloo Institute for Nanotechnology, University of Waterloo · Waterloo, ON, Canada

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Research Interests

Piezoelectric & triboelectric nanogenerators · Ferroelectric and Piezoelectric Materials · self-powered sensing · energy storage (Li-ion anodes, MXenes) · lead-free piezoelectric materials · DFT/first-principles modeling · machine-learning-enabled sensor systems · MEMS & cleanroom micro/nanofabrication.

Education

PhD, Electrical & Computer Engineering (Nanotechnology) (Sep 2022 – Fall 2026 (Expected))

University of Waterloo, Waterloo, ON, Canada

GPA 3.70/4.00 · Thesis: Next-Generation Nanogenerators for Sustainable Energy Solutions in Challenging Environments ·

Advisor: Prof. Dayan Ban

MSc, Physics (Sep 2018 – Feb 2021)

Sharif University of Technology, Tehran, Iran

GPA 3.82/4.00 · Thesis: Hybrid Nanostructures as Anode Material for Ionic Batteries

BSc, Physics (Condensed Matter) (Sep 2014 – Sep 2018)

Iran University of Science & Technology, Tehran, Iran

GPA 3.76/4.00 · Ranked #1 in cohort · Thesis: Simulating Surface Morphology of the Ballistic Deposition Model

Research Experience

Graduate Research Assistant (PhD Candidate) (Sep 2022 – Present)

University of Waterloo — Prof. Dayan Ban's Lab, Waterloo, ON

- Led three concurrent industry-collaborative R&D projects with the National Research Council of Canada (NRC) and Shimco North America, from planning through delivery — producing milestone reports, technical presentations, and peer-reviewed publications.
- Designed, fabricated, and characterized high-performance piezoelectric and triboelectric nanogenerators (PENGs/TENGs) for self-powered sensing, wearable electronics, and energy harvesting; results published in Advanced Functional Materials, Nano Energy, and Energy & Environmental Materials.
- Built a TENG-based acoustic sensor integrated with a Python machine-learning classifier for live sound monitoring and captioning; published in Advanced Sensor Research and presented at IEEE CCECE.
- Developed multi-layer cascaded PENG architectures enabling reliable operation under cryogenic and low-temperature conditions, expanding the device deployment envelope.
- Performed first-principles DFT simulations (Quantum ESPRESSO) on lead-free perovskites to investigate electronic structure and ferroelectric polarization, guiding material selection.
- Fabricated, simulated, and tested InGaN/GaN multiple-quantum-well PIN micro-photodetectors with on-chip grating couplers in the Quantum-Nano Fabrication & Characterization Facility (QNFCF) cleanroom, following a ~30-step multilayer process flow covering mesa definition, passivation, metallization, and top-cladding integration.
- Executed the full process-integration toolbox at QNFCF: PECVD and ALD dielectric deposition, e-beam evaporation and sputter metallization (Cr/Au, Al), hard-mask patterning, reactive-ion etching of metals, oxides, and ITO, lift-off, and wet processing (piranha cleaning, KOH treatment, wet metal etching) with photoresist and e-beam-resist workflows.
- Trained on and worked with electron-beam lithography, maskless laser lithography, plasma deposition and dry-etch cluster tools, resist spin coating, and solvent wet-bench processing at QNFCF.
- Investigated ferroelectric and Piezoelectric materials and their low-temperature phase transitions through experimental characterization and theoretical simulations using the Ferret MOOSE framework, focusing on phase-field modeling and domain evolution.

- Developed graphical user interfaces (GUIs) for lab-instrument control and automation, saving measurement time and equipment resources.
- Carried out full device fabrication in a class-controlled cleanroom: photolithography, thin-film deposition, electrode patterning, encapsulation, and electromechanical characterization.
- Designed and built custom motor-tapping test benches and a cryogenic test rig for reproducible benchmarking of energy-harvesting devices.
- Mentored co-op and Undergraduate Research Assistant (URA) students; coordinated parallel workflows across three collaborator-driven research streams.
- Co-authored a Nature Communications paper on perovskite piezocomposites; lab work featured by CTV News Kitchener and University of Waterloo News.

Graduate Research Assistant (MSc) (Feb 2021 – Feb 2022)

Sharif University of Technology, Tehran, Iran

- Led the battery research group: experimental planning, project workflows, and student-team coordination for lithium-ion anode development.
- Raised Li-ion anode performance ~50% and roughly doubled cycle life with a binder-free SnS_2 anode using $Ti_3C_2T_x$ MXene as a conductive binder.
- Synthesized SnS_2 nanostructured anodes; fabricated freestanding MXene and graphene-oxide papers; optimized binder systems (PEDOT:PSS, PVDF, CMC).
- Designed, built, and programmed a custom in-house glove box for oxygen- and moisture-sensitive cell fabrication, including sealing, atmosphere control, and instrumentation.
- Delivered industry milestones well ahead of schedule (~8 vs 18 months; 12 vs 24 months) while exceeding performance targets (~1.8× battery life).
- Developed a holey/porous-graphene anode (~20% gain) and self-healing MXene-based composite anodes; developed 2D-material transfer protocols for device integration.

Undergraduate Research Assistant (Nov 2019 – Feb 2021)

Sharif University of Technology, Tehran, Iran

- Synthesized SnS_2–graphene-oxide composites for Li-ion anodes; produced graphene oxide via Hummers' method and $Ti_3C_2T_x$ MXene via LiF/HCl etching.
- Industry collaboration (Parthian Battery Novin & Iran Nano Battery Network): contributed to a Sharif–industry R&D project developing high-performance Li-ion batteries with self-healing nanomaterials ($Ti_3C_2T_x$ MXenes) to improve cycling stability and mechanical resilience.
- Investigated MXene-based composite anodes and binder formulations to suppress capacity fade, mitigate volumetric expansion, and improve Coulombic efficiency.
- Coordinated experimental planning between the academic lab and industry partners; performed materials/device characterization (XRD, FTIR, SEM/EDX, TEM, I–V, C–V, CV, battery cycling).

Undergraduate Researcher (Feb 2018 – Aug 2018)

Iran University of Science & Technology, Tehran, Iran

- Implemented stochastic Monte Carlo simulations of the ballistic-deposition model to study non-equilibrium thin-film growth, surface-morphology evolution, and dynamic roughening on 1+1D and 2+1D substrates.
- Computed roughness scaling exponents and verified Family–Vicsek dynamic scaling, confirming Kardar–Parisi–Zhang (KPZ) universality-class behavior.
- Built foundational PVD and thin-film-growth understanding later applied to sputtering, CVD, and cleanroom fabrication; programmed simulation and visualization workflows in Python.

Publications (Google Scholar)

1. M. Haji Bagheri, A. A. Khan, M. F. Al Fattah, M. Al Madhoun, S. Shahzadi, A. Pan, M. S. Hasan, H. Wu, A. Aghigh, B. Bhawna, F. Légaré, V. Maheshwari, D. Ban, "Lone-Pair-Induced Lattice Distortion in Lead-Free Hybrid Perovskites for High-Performance Nanogenerators," Advanced Functional Materials, 2026.
2. M. F. Al Fattah, A. A. Khan, S. Khamgaonkar, M. Haji Bagheri, D. Ban, "Photoconductivity-Driven Quantum Efficiency Gain in Inorganic Ruddlesden-Popper Layered $Cs_2PbBr_2I_2$ Perovskite Photodetector for Visible Light Detection," Advanced Functional Materials, 2026, e22928.
3. M. Haji Bagheri, A. Rajabi-Abhari, O. Gibbs, P. Xi, A. A. Khan, G. Xiao, N. Yan, D. Ban, "Self-Powered Live Sound Monitoring and Classification via Triboelectric Nanogenerator and Machine Learning," 7th IEEE IFETC, Vancouver, 2025.
4. M. Haji Bagheri, A. A. Khan, M. Almadhoun, S. Shahzadi, D. Ban, "High-Performance Lead-Free Composite Piezoelectric Nanogenerator for Efficient Energy Harvesting and Self-Powered Applications," 7th Int. Conf. on NanoEnergy and NanoSystems (NENS 2025), Beijing.
5. M. Haji Bagheri, A. Rajabi-Abhari, O. Gibbs, P. Xi, A. A. Khan, F. Huang, Md. S. Hassan, N. Yan, D. Ban, "Sustainable Live Sound Monitoring and Classification System Enabled by a Triboelectric Nanogenerator and Machine Learning Techniques," Energy & Environmental Materials, 2025, e70044.
6. M. Haji Bagheri, E. Gu, A. A. Khan, Y. Zhang, G. Xiao, M. Nankali, P. Peng, P. Xi, D. Ban, "Machine Learning-Enabled Triboelectric Nanogenerator for Continuous Sound Monitoring and Captioning," Advanced Sensor Research, 2025, 2400156.
7. A. A. Khan, A. Mathur, L. Yin, M. Almadhoun, J. Yin, M. Haji Bagheri, M. F. Al Fattah, N. Yan, B. Zhao, V. Maheshwari, D. Ban, "Breaking Dielectric Dilemma via Polymer-Functionalized Perovskite Piezocomposite with Large Current-Density Output," Nature Communications, 2024, 15, 9511.
8. M. Haji Bagheri, A. Rajabi-Abhari, P. Li, A. A. Khan, C. Hao, N. R. Tanguy, D. Ban, L. Yu, N. Yan, "Self-Healable, Recyclable, and Mechanically Robust Vitrimer Composite for High-Performance Triboelectric Nanogenerators and Self-Powered Wireless Electronics," Nano Energy, 2024, 110306.
9. M. Haji Bagheri, J. Li, E. Gu, K. Habashy, M. M. Rana, A. A. Khan, Y. Zhang, G. Xiao, P. Xi, D. Ban, "A Data-Driven Environmental Sound Classification System with Acoustic Triboelectric Nanogenerators," IEEE CCECE, Kingston, 2024, pp. 881-886.
10. M. Haji Bagheri, A. A. Khan, S. Shahzadi, M. M. Rana, M. S. Hasan, D. Ban, "Advancements and Challenges in Molecular/Hybrid Perovskites for Piezoelectric Nanogenerator Application: A Comprehensive Review," Nano Energy, 2023, 109101.
11. M. S. Hasan, A. A. Khan, S. Shahzadi, M. Haji Bagheri, D. Ban, "Terahertz Radiation Detectors Using CMOS-Compatible SOI Substrates," Advanced Functional Materials, 2024, 2400313.
12. M. M. Rana, A. A. Khan, M. Haji Bagheri, D. Ban, "Doping Engineering Towards Stable Lead-Free Flexible Piezoelectric Nanogenerator for Self-Powered Wireless Electronics," 20th Canadian Semiconductor Science & Technology Conf. (CSSTC), 2024.
13. M. M. Rana, A. A. Khan, M. Haji Bagheri, D. Ban, "Chemically Functionalized High-Performance Lead-Free PENG for Flexible Electronics," IEEE 17th Dallas Circuits and Systems Conf. (DCAS), 2024.
14. A. A. Khan, M. M. Rana, M. Haji Bagheri, M. F. Al Fattah, M. S. Hassan, M. Almadhoun, D. Ban, "Unveiling Piezoelectric Property of a $CsPbCl_3$ Microcrystal for Flexible Nanogenerator Design," IEEE 17th DCAS, 2024.
15. M. Haji Bagheri, A. Esfandiar, M. Rastgoo, "Graphene/Tin(IV) Sulfide (SnS_2) Hybrid Nanostructures as High-Performance Anode Material for Rechargeable Li-Ion Batteries," Annual Physics Conf. of Iran 1400, oral presentation.
16. M. Haji Bagheri, A. Esfandiar, M. Rastgoo, "Investigation of Binder Effect on Electrochemical Performance of Graphene/SnS_2 Hybrid Nanostructures in Ultrafast Rechargeable Li-Ion Batteries," National Lithium Battery Conf. 2021, oral presentation.
17. Poster: "Self-Powered TENG for Environmental Sound Monitoring," Waterloo Nanotechnology Conference (WNC), 2024 — awarded Best Poster Prize.

Teaching Experience

- Invited Guest Lecture, Wilkes University, USA (April 2026): "Triboelectric Nanogenerator-Based Microphones Integrated with Machine Learning for Sustainable Sound Monitoring"; invited by Dr. Asif Abdullah Khan.
- Graduate Lab Assistant, University of Waterloo (ECE): Nano-Instrumentation Laboratory — MEMS device fabrication, simulation & characterization (Prof. Hany Aziz, B. Sadeghimakki), Winter 2023 & 2024.
- Graduate Lab Assistant, University of Waterloo (ECE): Microfabrication & Thin-Film Technology Laboratory — electronic-device simulation & fabrication (Prof. B. Sadeghimakki), Fall 2023.
- Graduate Teaching Assistant, University of Waterloo (ECE): Characterization of Nanomaterials (Prof. Guo-Xing Miao), Spring 2023.

- Graduate Teaching Assistant, University of Waterloo (ECE): Advanced Calculus & Numerical Methods I & II (Prof. R. Thompson, P. Neathway), 2024–2026; Numerical Methods (Prof. D. Hardner), Winter 2025 & 2026.
- Graduate Lab Assistant, Sharif University of Technology (Physics): Advanced Physics Laboratory for MSc students (Prof. A. Irajizad & N. Taghavinia), 2020–2022; Nano-Scale Physics Laboratory (Prof. A. Esfandiar), 2021–2022.

Awards & Honors

- Best Poster Prize — Waterloo Nanotechnology Conference (WNC), University of Waterloo, 2024.
- WIN Nanofellowship Award — Waterloo Institute for Nanotechnology.
- Graduate Student Research Dissemination (GSRD) Award — University of Waterloo.
- Faculty of Engineering (FoE) Award — University of Waterloo (×3).
- Research Publication Award & Publication Excellence Award — University of Waterloo (Prof. Dayan Ban's lab).
- Direct Admission to MSc, Sharif University of Technology (top BSc rank).
- Ranked #1 among BSc Physics students, Iran University of Science & Technology.
- CTV News Kitchener interview & University of Waterloo News feature on Nature Communications PENG publication.

Technical Skills

Device Fabrication: PENG/TENG and photodetector fabrication; III-nitride (InGaN/GaN MQW) device processing; Li-ion cell assembly; photolithography; electron-beam lithography; maskless laser lithography; thin-film deposition (sputtering, e-beam evaporation, PECVD, ALD, CVD, PVD, CVT); resist spin coating (photo & e-beam resists); reactive-ion etching (RIE); lift-off; wet processing (piranha, KOH, solvent wet bench); electrospinning; spray & vacuum-filtration coating; MEMS/CMOS/diode/MOSFET fabrication; bonding; class-controlled cleanroom operation (incl. QNFCF).

Characterization: XRD; FTIR; SEM/EDX; TEM; AFM; optical microscopy; I–V; C–V; EIS; cyclic voltammetry; galvanostatic battery cycling; electromechanical & electroacoustic testing.

Simulation & Modeling: Quantum ESPRESSO (DFT, electronic structure, ferroelectric polarization); COMSOL Multiphysics; Ferret/MOOSE phase-field modeling of ferroelectric materials, domain evolution, and temperature-dependent phase transitions; Ansys Lumerical; Monte Carlo / stochastic thin-film growth (KPZ-class); MATLAB.

Programming & Data: Python (signal processing, ML classification on sensor data); MATLAB; Origin Pro; LaTeX; Microsoft Office.

Custom Instrumentation: Motor-tapping energy-harvester test benches; cryogenic / low-temperature test rigs; oxygen-controlled glove boxes.

Leadership & Collaboration: Industry/government project planning; milestone reporting; technical presentations (NRC, Shimco); mentoring of co-op and undergraduate research assistants.

Certification: Certified Radiation Protection Professional — Non-Destructive Testing (NDT), Medical & Industrial.